

Mod 4, Task 3_ Critical Annotation, Bender
Critical Annotation Form:

ANNOTATION 1 — COMPELLING Claim (quote or paraphrase with section number):

Section 1: Bender et al. argue that the environmental and financial costs of ever-larger language models should be weighed before assuming that scale is progress, especially because the benefits and harms are not distributed equally.

ANNOTATION 2 — COMPELLING Claim (quote or paraphrase with section number):

Section 1: The authors argue that large, web-derived datasets can overrepresent dominant or hegemonic viewpoints and create documentation debt when the data are too large to understand, curate, or document after the fact.

ANNOTATION 3 — COMPELLING Claim (quote or paraphrase with section number):

Section 6: The paper argues that when language models reproduce biased or abusive patterns from training data, their fluent output can amplify stereotypes, microaggressions, and discriminatory associations at scale.

ANNOTATION 4 — CHALLENGE Claim (quote or paraphrase with section number):

Section 6.1: Bender et al. claim that text generated by a language model is not grounded in communicative intent, any model of the world, or any model of the reader's state of mind; the model is therefore manipulating form rather than meaning.

Argument type (A–E): Why analytically compelling (1–2 sentences):

B. Environmental argument. This is analytically compelling because it connects technical model scaling to resource use and equity. It asks not only whether a model performs better, but who pays the environmental and financial cost of that performance.

Argument type (A–E): Why analytically compelling (1–2 sentences):

A. Documentation argument. This is strong because it identifies a practical accountability problem: if developers cannot explain what is in the training data, downstream users cannot adequately assess bias, representativeness, or risk.

Argument type (A–E): Why analytically compelling (1–2 sentences):

C. Bias encoding argument. This is analytically sound because the harm is not only that bias exists in the data, but that fluent synthetic text can make biased associations easier to spread, normalize, and reuse in future systems.

Argument type (A–E): Challenge / counter-argument (2–3 sentences):

D. Epistemic/comprehension argument. A scope condition is needed: Wei et al. (2022) and emergent-abilities research suggest that large models can sometimes perform reasoning-like tasks that are not obvious in smaller models, which complicates the blanket claim that they have no world-model-like internal structure. This does not prove human understanding or communicative intent, but it weakens the strongest version of the claim if it is read as only parroting with no meaningful internal representation.